

Aashish Awadhani

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SUMMARY

Machine Learning Engineer with expertise in real-time computer vision, OCR systems, and LLM-based AI applications. Skilled in deploying optimized models, accelerating inference, quantization, and pruning, and building end-to-end ML pipelines. Strong engineering background with the ability to deliver full-stack systems, scalable APIs, and production-ready AI features.

EXPERIENCE

Hydhan

Feb 2026 - Present

Software Engineer, AI Operations

California

- Developed an end-to-end automation engine that integrates Vision Transformers for hardware classification and Large Language Models for autonomous order synthesis.
- Engineered a high-throughput middleware bridge using Python (FastAPI) and Celery/Redis to synchronize real-time data across disparate third-party (ERP) platforms.
- Integrated an Intelligent Document Processing (IDP) pipeline using Hugging Face Transformers for Named Entity Recognition (NER) on unstructured technical manifests and purchase orders.

CalSys Lab, Cal Poly Pomona

Sep 2025 - Present

AI/ML Research Assistant

California

- Built and benchmarked 4 OCR pipelines (CNN, YOLO-based detection, CRNN-CTC, Transformer OCR) for CAPTCHA recognition, improving sequence accuracy across noisy inputs.
- Developed character recognition models using PyTorch/TensorFlow, utilizing CNNs and Vision Transformers (ViTs) for robustness under distortion.
- Improved handling of overlapping characters using a hybrid YOLO + CRNN architecture, reducing segmentation errors across complex datasets.

Autonomous Vehicle Lab, Cal Poly Pomona

May 2024 - May 2025

Machine Learning Engineer

California

- Engineered a real-time object detection stack using PyTorch + TensorRT on Jetson AGX Orin, achieving 35–40% lower latency and more stable FPS in field tests.
- Integrated YOLOv8 with ROS2 to enhance autonomous navigation using multi-sensor perception (RGB, LiDAR, depth).
- Performed model compression via quantization, pruning, and ONNX/TensorRT optimization, enabling reliable deployment under embedded hardware constraints.

EDUCATION

California State Polytechnic University, Pomona

May 2025

Master of Science, Computer Science

PROJECTS

Real-Time Chat Application | MongoDB, Express.js, React, Node.js

- Built a responsive chat platform using MongoDB, Express.js, React, Node.js, and Socket.io with JWT-based authentication and global state via Zustand.
- Deployed real-time messaging infrastructure with WebSocket handling and clean UI using Tailwind CSS.

Stock Price Prediction with XGBoost | TensorFlow, Python, JavaScript

- Developed a time-series forecasting model with XGBoost, achieving 94% accuracy using advanced feature engineering (RSI, MACD, Bollinger Bands).
- Built a complete ML pipeline in Python with cross-validation and visualized trends using Matplotlib.

CERTIFICATION

- **Rapid Application Development with Large Language Models (LLMs) : NVIDIA Deep Learning Institute**
 - * Built LLM-based applications using Transformers, LangChain, and vector databases.
 - * Developed text-generation, RAG, and semantic search pipelines for production-style use cases.

SKILLS

- **Machine Learning & AI:** PyTorch, TensorFlow, TensorRT, YOLO, OpenCV, CNN, RNN, Vision Transformers, Transformers, ONNX, Quantization, Pruning, CUDA, MLflow, Large Language Models, Hugging Face, LangChain, RAG, Vector DB, Natural Language Processing, Generative AI, Diffusion Models, AI Agents, OpenAI API, LLMops
- **Software Engineering & Systems:** Python, C++, Java, JavaScript, Node.js, Express.js, Rest APIs, Docker, Git, Linux, ROS2, Fast API, MySQL, PostgreSQL, MongoDB, Redis, CI/CD
- **Cloud & Tools:** AWS, GCP, Azure, GitHub Actions, PowerBI, Tableau